



**WORK BREAKDOWN STRUCTURE (WBS)
WAH4HOSPITAL (WAH4H)**

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INTRODUCTION

This Work Breakdown Structure (WBS) defines all deliverables and work packages required to complete the WAH4H Hospital Information System project. The current deliverable is a demonstration version — a fully functional proof-of-concept intended for presentation to government officials (governors and mayors) to secure institutional buy-in and funding. A production-ready version will be developed in a subsequent phase following the demonstration.

The project scope covers all 11 modules (PBL1 and PBL2), cross-system interoperability with WAH4P and WAH4PC, infrastructure deployment on WAH-managed Proxmox via Docker, complete turnover documentation, and formal project closure with sign-off from Ms. Anna Katrina Yturralde (Health Program Supervisor) and Sir Pey Canlas (Executive Director, WAH).

OUTLINE VIEW

The following outline represents the full decomposition of project work:

WBS ID	Phase / Deliverable Group
1.0	WAH4H Hospital Information System Project
1.1	Project Management and Planning
1.2	System Design and Compliance Architecture
1.3	Infrastructure Preparation and Deployment Pipeline
1.4	PBL1 Module Development and Validation
1.5	PBL2 Module Development and Validation
1.6	Cross-System Interoperability
1.7	Security Implementation
1.8	User Acceptance Testing and Defect Resolution
1.9	Turnover Documentation
1.10	System Deployment and Handoff
1.11	Project Closure

HIERARCHICAL STRUCTURE

The table below presents the WBS in an indented hierarchical format, showing parent-child relationships across all decomposition levels.

Phase	Deliverables / Work Packages
1.1 Project Management and Planning	Project Charter • Stakeholder Register and RACI Matrix • Scope Management Plan • Schedule Baseline and Milestone Tracking Record • Risk and Issue Register • Communications



	Management Plan • Human Resource Plan • Work Breakdown Structure (WBS)
1.2 System Design and Compliance Architecture	System Requirements Specification (SRS) • Use Case Document (UCD) • System Design Document (SDD) • Data Flow Diagrams (L0–L2) • Deployment Architecture Diagram • PH Core FHIR R4 Compliance Matrix • Data Privacy and UHC Compliance Review (RA 10173 and RA 11223)
1.3 Infrastructure Preparation and Deployment Pipeline	Proxmox VE Installation and Configuration • VM Structure Definition (Local / Sandbox / Production) • Docker Runtime Installation and Compatibility Verification • Docker Compose Configuration (Frontend / Backend / Database) • Cloudflare Tunnel Configuration • Fail2Ban Firewall Configuration • Three-Tier Pipeline Validation (Local → Sandbox → Production)
1.4 PBL1 Module Development and Validation	Accounts Module — RBAC (7 Roles), JWT Authentication, Role-Specific Dashboards • Patients Module — PH Core FHIR R4 Patient Resource, PSGC, PhilHealth Number • Admission Module — FHIR Encounter, ICD-10-PH Admitting Diagnosis, Ward/Room Assignment, Insurance Linkage • Monitoring Module — FHIR Observation, Vital Signs (Temp, BP, PR, RR, O2 Sat), Nursing Notes, Physician Orders, I&O Tracking • Discharge Module — FHIR Encounter Discharge Disposition, ICD-10-PH, Discharge Summary, Follow-Up, Clearance Generation • Billing Module — FHIR Invoice/ChargeItem, PhilHealth Deductions, HMO Deductions, Patient Payable, OR Generation • Pharmacy Module — FHIR MedicationRequest/Administration, MIMS PH Drug Coding, Prescription Processing, Stock Management, Low-Stock Alerts • Laboratory Module — FHIR DiagnosticReport/Observation, LOINC-Coded Tests, Specimen Tracking, Result Entry, Result Release
1.5 PBL2 Module Development and Validation	Admin Module — FHIR Organization/Location/PractitionerRole, Facility Config, Staff Management, Department/Ward Setup, Audit Logs • Appointments Module — FHIR Appointment, Booking/Rescheduling/Cancellation, Calendar Views per Physician/Department, Notifications • eClaims Module — FHIR Claim/ClaimResponse, CF1/CF2/CF4 Forms, RVS Coding, Batch Submission, Remittance Advice, Payment Reconciliation
1.6 Cross-System Interoperability	WAH4P Integration Technical Specification • WAH4P Patient and Clinical Data Synchronization • WAH4PC Integration Technical Specification • WAH4PC Data Synchronization • Cross-Platform FHIR Resource Reference Validation



1.7 Security Implementation	JWT Authentication (Token Expiry and Refresh) • RBAC Enforcement Across All 11 Modules • Audit Logging — All Clinical and Financial Transactions • Data Encryption at Rest and In Transit (RA 10173) • Cloudflare Tunnel and Fail2Ban Deployment Validation • Environment Separation Enforcement (Local / Sandbox / Production)
1.8 User Acceptance Testing and Defect Resolution	UAT Test Case Preparation (All 11 Modules) • UAT Execution with WAH Stakeholders • UAT Defect Log and Issue Resolution Record • Re-Testing After Defect Resolution • UAT Sign-Off Sheets — Ms. Anna Katrina Yturalde (Health Program Supervisor)
1.9 Turnover Documentation	Training Manual — Module Workflows, User Roles, Common Scenarios, Troubleshooting • End-User Manual — Role-Specific Guides (Doctor, Nurse, Pharmacist, Lab Tech, Billing Staff, Admin, Hospital Admin) • Technical Manual — Architecture, Proxmox/Docker Setup, Server Config, DB Management, Backup and Restore, Maintenance • UAT Documentation Package — Test Cases, Results, Defect Logs, Sign-Off Sheets • Source Code and Repository Turnover — Repo Access, Docker Compose Files, Env Configs, Deployment Scripts
1.10 System Deployment and Handoff	Production VM Provisioning on Proxmox (Sir Robert Michael Martinez) • Final Container Stack Deployment to Production • Production System Initialization Validation • Sandbox-to-Production Environment Parity Verification • WAH Staff Training Delivery • Formal Handoff — Ms. Anna Katrina Yturalde Acceptance then Sir Pey Canlas Final Sign-Off
1.11 Project Closure	Final Academic Demonstration • Formal Acceptance Sign-Off (Ms. Anna Katrina Yturalde then Sir Pey Canlas) • Project Closeout Report • Lessons Learned Document

TABULAR VIEW

The tabular view lists every WBS ID and its corresponding work package or deliverable.

WBS ID	Work Package / Deliverable
1.0	WAH4H Hospital Information System Project
1.1	Project Management and Planning
1.1.1	Project Charter
1.1.2	Stakeholder Register and RACI Matrix
1.1.3	Scope Management Plan



1.1.4	Schedule Baseline and Milestone Tracking Record
1.1.5	Risk and Issue Register
1.1.6	Communications Management Plan
1.1.7	Human Resource Plan
1.1.8	Work Breakdown Structure (WBS)
1.2	System Design and Compliance Architecture
1.2.1	System Requirements Specification (SRS)
1.2.2	Use Case Document (UCD)
1.2.3	System Design Document (SDD)
1.2.4	Data Flow Diagrams (L0–L2)
1.2.5	Deployment Architecture Diagram
1.2.6	PH Core FHIR R4 Compliance Matrix
1.2.7	Data Privacy and UHC Compliance Review (RA 10173 and RA 11223)
1.3	Infrastructure Preparation and Deployment Pipeline
1.3.1	Proxmox VE Installation and Configuration
1.3.2	VM Structure Definition (Local / Sandbox / Production)
1.3.3	Docker Runtime Installation and Compatibility Verification
1.3.4	Docker Compose Configuration (Frontend / Backend / Database)
1.3.5	Cloudflare Tunnel Configuration
1.3.6	Fail2Ban Firewall Configuration
1.3.7	Three-Tier Pipeline Validation (Local → Sandbox → Production)
1.4	PBL1 Module Development and Validation
1.4.1	Accounts Module
1.4.1.1	RBAC Implementation (7 Roles: Doctor, Nurse, Pharmacist, Lab Tech, Billing Staff, Admin, Hospital Admin)
1.4.1.2	JWT Authentication (Token Expiry and Refresh)
1.4.1.3	Role-Specific Dashboards and Access Controls
1.4.2	Patients Module
1.4.2.1	PH Core FHIR R4 Patient Resource Implementation
1.4.2.2	PSGC Address Coding Integration
1.4.2.3	PhilHealth Number, Civil Status, Blood Type, Emergency Contact
1.4.3	Admission Module
1.4.3.1	FHIR Encounter Resource Mapping
1.4.3.2	ICD-10-PH Admitting Diagnosis Integration
1.4.3.3	Ward/Room Assignment, Attending Physician, Insurance Linkage
1.4.4	Monitoring Module
1.4.4.1	FHIR Observation — Vital Signs (Temp, BP, PR, RR, O2 Sat)
1.4.4.2	Nursing Notes and Physician Orders



1.4.4.3	Intake and Output Tracking
1.4.5	Discharge Module
1.4.5.1	FHIR Encounter Discharge Disposition
1.4.5.2	ICD-10-PH Discharge Diagnosis
1.4.5.3	Discharge Summary, Follow-Up Instructions, Clearance Generation
1.4.6	Billing Module
1.4.6.1	FHIR Invoice and ChargeItem Resources
1.4.6.2	PhilHealth Benefit Deductions and HMO Deductions
1.4.6.3	Patient Payable Computation and Official Receipt Generation
1.4.7	Pharmacy Module
1.4.7.1	FHIR MedicationRequest and MedicationAdministration
1.4.7.2	MIMS PH Drug Coding Integration
1.4.7.3	Prescription Processing, Stock Management, Low-Stock Alerts
1.4.8	Laboratory Module
1.4.8.1	FHIR DiagnosticReport and Observation Resources
1.4.8.2	LOINC-Coded Test Integration
1.4.8.3	Specimen Tracking, Result Entry, and Result Release to Attending Physician
1.5	PBL2 Module Development and Validation
1.5.1	Admin Module
1.5.1.1	FHIR Organization, Location, and PractitionerRole Resources
1.5.1.2	Facility Configuration, Staff Management, Department/Ward Setup, System Settings
1.5.1.3	Audit Log Implementation
1.5.2	Appointments Module
1.5.2.1	FHIR Appointment Resource Implementation
1.5.2.2	Booking, Rescheduling, and Cancellation Workflows
1.5.2.3	Calendar Views per Physician and Department
1.5.2.4	Notifications and Reminder Support
1.5.3	eClaims Module
1.5.3.1	FHIR Claim and ClaimResponse Resources
1.5.3.2	CF1/CF2/CF4 Form Generation
1.5.3.3	RVS Procedure Coding Integration
1.5.3.4	eClaims Batch Submission
1.5.3.5	Remittance Advice Processing and Payment Reconciliation
1.6	Cross-System Interoperability
1.6.1	WAH4P Integration Technical Specification
1.6.2	WAH4P Patient and Clinical Data Synchronization
1.6.3	WAH4PC Integration Technical Specification



1.6.4	WAH4PC Data Synchronization
1.6.5	Cross-Platform FHIR Resource Reference Validation
1.7	Security Implementation
1.7.1	JWT Authentication Implementation (Token Expiry and Refresh)
1.7.2	RBAC Enforcement Across All 11 Modules
1.7.3	Audit Logging — All Clinical and Financial Transactions
1.7.4	Data Encryption at Rest and In Transit (RA 10173)
1.7.5	Cloudflare Tunnel and Fail2Ban Deployment Validation
1.7.6	Environment Separation Enforcement (Local / Sandbox / Production)
1.8	User Acceptance Testing and Defect Resolution
1.8.1	UAT Test Case Preparation (All 11 Modules)
1.8.2	UAT Execution with WAH Stakeholders
1.8.3	UAT Defect Log and Issue Resolution Record
1.8.4	Re-Testing After Defect Resolution
1.8.5	UAT Sign-Off Sheets — Ms. Anna Katrina Yturralde (Health Program Supervisor)
1.9	Turnover Documentation
1.9.1	Training Manual
1.9.2	End-User Manual
1.9.3	Technical Manual
1.9.4	UAT Documentation Package
1.9.5	Source Code and Repository Turnover Package
1.10	System Deployment and Handoff
1.10.1	Production VM Provisioning on Proxmox (Sir Robert Michael Martinez)
1.10.2	Final Container Stack Deployment to Production
1.10.3	Production System Initialization Validation
1.10.4	Sandbox-to-Production Environment Parity Verification
1.10.5	WAH Staff Training Delivery
1.10.6	Formal Handoff — Ms. Anna Katrina Yturralde Acceptance then Sir Pey Canlas Final Sign-Off
1.11	Project Closure
1.11.1	Final Academic Demonstration
1.11.2	Formal Acceptance Sign-Off (Ms. Anna Katrina Yturralde then Sir Pey Canlas)
1.11.3	Project Closeout Report
1.11.4	Lessons Learned Document

TREE STRUCTURE VIEW

The tree structure below uses ASCII notation to represent parent-child relationships across the full WBS hierarchy.



- 1.0 WAH4H Hospital Information System Project**
- 1.1 Project Management and Planning**
 - 1.1.1 Project Charter
 - 1.1.2 Stakeholder Register and RACI Matrix
 - 1.1.3 Scope Management Plan
 - 1.1.4 Schedule Baseline and Milestone Tracking Record
 - 1.1.5 Risk and Issue Register
 - 1.1.6 Communications Management Plan
 - 1.1.7 Human Resource Plan
 - 1.1.8 Work Breakdown Structure (WBS)
- 1.2 System Design and Compliance Architecture**
 - 1.2.1 System Requirements Specification (SRS)
 - 1.2.2 Use Case Document (UCD)
 - 1.2.3 System Design Document (SDD)
 - 1.2.4 Data Flow Diagrams (L0-L2)
 - 1.2.5 Deployment Architecture Diagram
 - 1.2.6 PH Core FHIR R4 Compliance Matrix
 - 1.2.7 Data Privacy and UHC Compliance Review (RA 10173 and RA 11223)
- 1.3 Infrastructure Preparation and Deployment Pipeline**
 - 1.3.1 Proxmox VE Installation and Configuration
 - 1.3.2 VM Structure Definition (Local / Sandbox / Production)
 - 1.3.3 Docker Runtime Installation and Compatibility Verification
 - 1.3.4 Docker Compose Configuration (Frontend / Backend / Database)
 - 1.3.5 Cloudflare Tunnel Configuration
 - 1.3.6 Fail2Ban Firewall Configuration
 - 1.3.7 Three-Tier Pipeline Validation (Local to Sandbox to Production)
- 1.4 PBL1 Module Development and Validation**
 - 1.4.1 Accounts Module
 - 1.4.2 Patients Module
 - 1.4.3 Admission Module
 - 1.4.4 Monitoring Module
 - 1.4.5 Discharge Module
 - 1.4.6 Billing Module
 - 1.4.7 Pharmacy Module
 - 1.4.8 Laboratory Module
- 1.5 PBL2 Module Development and Validation**
 - 1.5.1 Admin Module
 - 1.5.2 Appointments Module
 - 1.5.3 eClaims Module
- 1.6 Cross-System Interoperability**
 - 1.6.1 WAH4P Integration Technical Specification
 - 1.6.2 WAH4P Patient and Clinical Data Synchronization
 - 1.6.3 WAH4PC Integration Technical Specification
 - 1.6.4 WAH4PC Data Synchronization
 - 1.6.5 Cross-Platform FHIR Resource Reference Validation
- 1.7 Security Implementation**
 - 1.7.1 JWT Authentication Implementation
 - 1.7.2 RBAC Enforcement Across All 11 Modules
 - 1.7.3 Audit Logging - All Clinical and Financial Transactions
 - 1.7.4 Data Encryption at Rest and In Transit (RA 10173)



1.7.5 Cloudflare Tunnel and Fail2Ban Deployment Validation
1.7.6 Environment Separation Enforcement (Local / Sandbox / Production)

1.8 User Acceptance Testing and Defect Resolution

1.8.1 UAT Test Case Preparation (All 11 Modules)
1.8.2 UAT Execution with WAH Stakeholders
1.8.3 UAT Defect Log and Issue Resolution Record
1.8.4 Re-Testing After Defect Resolution
1.8.5 UAT Sign-Off Sheets - Ms. Anna Katrina Yturralde (Health Program Supervisor)

1.9 Turnover Documentation

1.9.1 Training Manual
1.9.2 End-User Manual
1.9.3 Technical Manual
1.9.4 UAT Documentation Package
1.9.5 Source Code and Repository Turnover Package

1.10 System Deployment and Handoff

1.10.1 Production VM Provisioning on Proxmox
1.10.2 Final Container Stack Deployment to Production
1.10.3 Production System Initialization Validation
1.10.4 Sandbox-to-Production Environment Parity Verification
1.10.5 WAH Staff Training Delivery
1.10.6 Formal Handoff - Ms. Anna Katrina Yturralde Acceptance then Sir Pey Canlas Final Sign-Off

1.11 Project Closure

1.11.1 Final Academic Demonstration
1.11.2 Formal Acceptance Sign-Off - Ms. Anna Katrina Yturralde then Sir Pey Canlas
1.11.3 Project Closeout Report
1.11.4 Lessons Learned Document

WBS FLOWCHART

The figure below presents the simplified flowchart of the WAH4H Work Breakdown Structure. This version shows only the major WBS categories and primary work packages, without the detailed sub-sections, so the diagram remains readable in the main document. The full breakdown, including the more detailed sub-work packages and descriptions, is provided in the WBS Dictionary section.

WBS DICTIONARY

The WBS Dictionary provides description, deliverable, and ownership detail for each top-level work package.

ID	Work Package	Description	Deliverable	Owner
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1.1	Project Management and Planning	All PMBOK planning artifacts establishing baseline scope, schedule, cost, risk, and communications.	Project Charter, Stakeholder Register, RACI, Risk Register, Comms Plan, HRP, WBS	Elijah Josh R. Quibin (PM)
1.2	System Design and Compliance Architecture	System requirements documentation, FHIR R4 compliance mapping, and regulatory compliance review (RA 10173, RA 11223). Compliance must be established at schema level before feature development.	SRS, SDD, UCD, DFDs, Deployment Diagram, FHIR R4 Compliance Matrix, Compliance Review	Jhon Lloyd Nicolas (Lead Developer / DevOps)
1.3	Infrastructure Preparation and Deployment Pipeline	Proxmox VE setup, VM provisioning across three tiers, Docker installation and configuration, Cloudflare Tunnel, Fail2Ban, and full pipeline validation.	Validated Local → Sandbox → Production pipeline on Proxmox	Mariyah Vanna Monique C. Chavez (QA)
1.4	PBL1 Module Development and Validation	Development and validation of all 8 core clinical modules with full PH Core FHIR R4 compliance and Philippine standards integration (ICD-10-PH, LOINC, MIMS PH, PSGC, SNOMED CT).	8 functional modules: Accounts, Patients, Admission, Monitoring, Discharge, Billing, Pharmacy, Laboratory	Jhon Lloyd Nicolas (Lead Developer / DevOps)
1.5	PBL2 Module Development and Validation	Development and validation of the 3 additional modules required in the final build: Admin, Appointments, and eClaims (including CF1/CF2/CF4 forms, RVS coding, and PhilHealth	Admin, Appointments, eClaims modules	Jhon Lloyd Nicolas (Lead Developer / DevOps)



		eClaims batch submission).		
1.6	Cross-System Interoperability	Technical specification and implementation of data synchronization between WAH4H and WAH4P/WAH4PC platforms. Sync events, data fields, and triggers remain in-scope but dependency-gated pending WAH technical specification from Sir Robert Michael Martinez before final validation.	WAH4P and WAH4PC sync integrations, validated FHIR cross-platform references	Jhon Lloyd Nicolas / Sir Robert Michael Martinez
1.7	Security Implementation	Application and infrastructure-layer security covering RBAC (7 roles), JWT, audit logging for all clinical and financial transactions, data encryption, Cloudflare Tunnel, and Fail2Ban per RA 10173.	Security-compliant system across all 11 modules and infrastructure	Jhon Lloyd Nicolas / Mariyah Vanna Monique C. Chavez (QA)
1.8	User Acceptance Testing and Defect Resolution	UAT preparation, execution, defect tracking, resolution, and re-testing across all 11 modules (PBL1 and PBL2). Formal sign-off from Ms. Anna Katrina Yturralde (Health Program Supervisor) required before handoff.	Completed UAT package with sign-off sheets	Elijah Josh R. Quibin (PM) / Jhon Lloyd Nicolas
1.9	Turnover Documentation	Three non-negotiable closure deliverables (Training Manual, End-User Manual, Technical Manual) plus UAT documentation package	Training Manual, End-User Manual, Technical Manual, UAT Docs, Source Code Package	All — Team Pseudoers



		and full source code turnover. Required for formal project closure per Sir Pey Canlas.		
1.10	System Deployment and Handoff	Production VM deployment to WAH-managed Proxmox infrastructure (Sir Robert Michael Martinez), final validation, WAH staff training delivery, and formal handoff per sign-off chain: Anna Katrina Yturralde approves first, then Sir Pey Canlas provides final sign-off.	Deployed production system, trained WAH staff, completed handoff	Mariyah Vanna Monique C. Chavez (QA) / Sir Robert Michael Martinez
1.11	Project Closure	Final academic demonstration, formal acceptance sign-off (Ms. Anna Katrina Yturralde then Sir Pey Canlas), project closeout reporting, and lessons learned documentation.	Project Closeout Report, Lessons Learned, Signed Acceptance Forms	Elijah Josh R. Quibin (PM)

GLOSSARY OF TERMS

Definitions for all terms, standards, and acronyms used throughout this document.

Term	Definition
WBS	Work Breakdown Structure. A hierarchical decomposition of all project deliverables and work packages.
PBL1	Product Backlog Level 1. The 8 core WAH4H modules: Accounts, Patients, Admission, Monitoring, Discharge, Billing, Pharmacy, Laboratory. UAT already completed.
PBL2	Product Backlog Level 2. The 3 additional modules required in the final build: Admin, Appointments, eClaims.
Demo Version	The current deliverable scope — a fully functional proof-of-concept for government stakeholder presentation. A production-ready version is a separate subsequent engagement.



FHIR R4	Fast Healthcare Interoperability Resources Release 4. The international standard for health data exchange.
PH Core FHIR R4	Philippine adaptation of the FHIR R4 Implementation Guide. Governs all clinical resource structures in WAH4H.
RBAC	Role-Based Access Control. Access permissions enforced per the 7 defined hospital roles: Doctor, Nurse, Pharmacist, Lab Tech, Billing Staff, Admin, Hospital Admin.
JWT	JSON Web Token. Stateless authentication mechanism for all WAH4H API endpoints. Tokens expire and are refreshed securely.
ICD-10-PH	International Classification of Diseases, Philippine adaptation. Used for diagnosis coding in Admission, Monitoring, Discharge, and eClaims.
LOINC	Logical Observation Identifiers Names and Codes. Used for laboratory test and clinical observation coding.
SNOMED CT	Systematized Nomenclature of Medicine Clinical Terms. Clinical terminology for conditions, procedures, and findings in Monitoring and Discharge modules.
MIMS PH	Philippine drug reference system. Applied to medication coding in the Pharmacy module.
RVS	Relative Value Scale. Procedure cost coding system used in PhilHealth eClaims.
PSGC	Philippine Standard Geographic Code. Standardized coding for all patient and facility address fields.
NHFR	National Health Facility Registry. Required for facility identification and eClaims compliance. WAH must register independently — out of Team Pseudoers' scope.
RA 10173	Republic Act 10173 — Data Privacy Act of 2012. Governs all patient data handling, encryption, access logging, and audit trails.
RA 11223	Republic Act 11223 — Universal Health Care Act. Governs PhilHealth interoperability requirements relevant to the eClaims and Billing modules.
Proxmox VE	Proxmox Virtual Environment. On-premise hypervisor used for WAH4H deployment. Managed by Sir Robert Michael Martinez on the WAH side.
Three-Tier Pipeline	Local (development) → Sandbox (staging/testing) → Production. No direct deployment to production without passing sandbox validation.
UAT	User Acceptance Testing. Formal testing conducted with WAH stakeholders. Sign-off authority: Ms. Anna Katrina Yturalde (Health Program Supervisor).
Sign-Off Chain	Ms. Anna Katrina Yturalde (Health Program Supervisor) approves system turnover first. Sir Pey Canlas (Executive Director) provides final sign-off only after Ms. Anna Katrina Yturalde's approval.



WAH4P	WAH patient-facing platform. WAH4H must sync patient and clinical data with this platform.
WAH4PC	WAH community/portal platform. WAH4H must integrate and sync data with this platform.

